

Cloud Computing Fundamentals

Learning Walkthrough

Room Name	Cloud Computing Fundamentals
Date	2026, June
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Overview

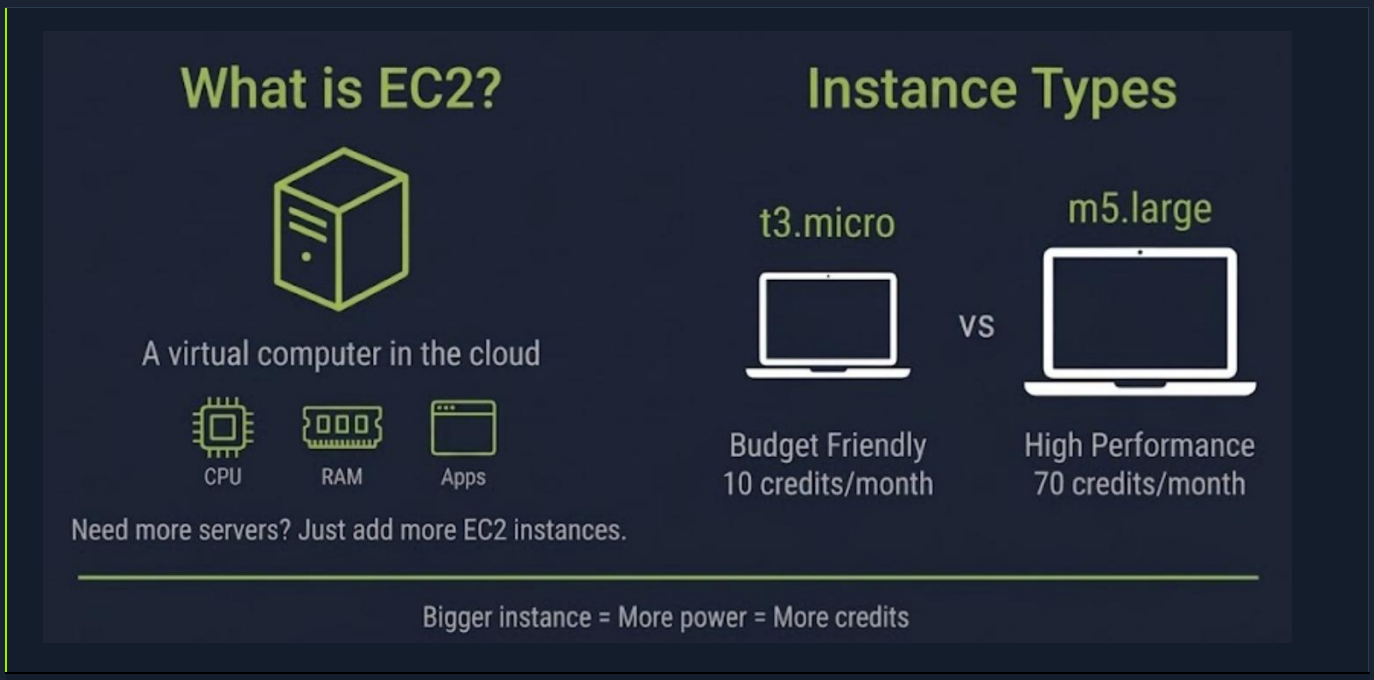
Hey! I just completed the Cloud Computing Fundamentals room on TryHackMe and wanted to write this up in a simple way that makes sense. This room is not about exploitation. It is about understanding what cloud computing is, how virtual machines work in the cloud, how to deploy them, and how billing works based on what you run and what you stop. I will walk through every part and explain what I understood from each one because the concepts here are foundational before jumping into anything cloud security related later.

01 Basic Cloud Terminology

Before touching anything practical the room starts by explaining the core terms. Let me break down what I learned here.

EC2 stands for Elastic Compute Cloud. Think of it as a virtual computer living in the cloud. Just like a real computer it has CPU and memory and can run applications. Whenever you need an extra server, you add an EC2 instance. You are basically adding a computer to your environment without buying physical hardware.

Instance types are the different sizes and power levels of those computers. For example, t2, t3, and m5 are different families. You choose based on what you need. A bigger instance like m5.large is like a high RAM laptop for heavy work. A smaller one like t3.micro is budget friendly for light tasks. The pricing scales with power.



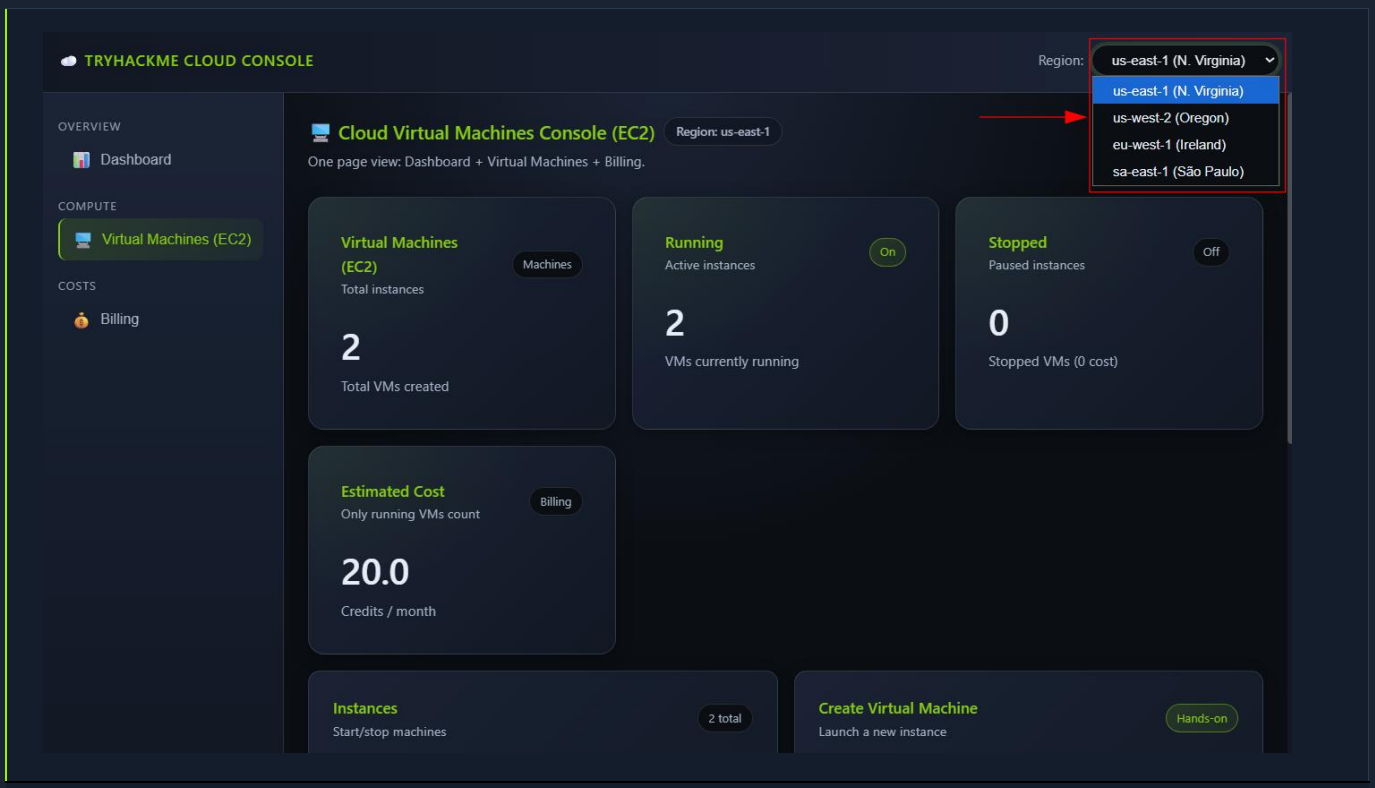
[*Fig: EC2 and Instance type in visually*]

> **NOTE**
>
> EC2 instance types follow a naming pattern. The letter before the dot is the family (t for general purpose, m for memory balanced). The number after it is the generation. The word after the dot is the size. t3.micro is third generation general purpose micro. m5.large is fifth generation memory optimized large. Understanding this pattern helps you pick the right size without guessing.

02 Deploying Environment

Picking a Region

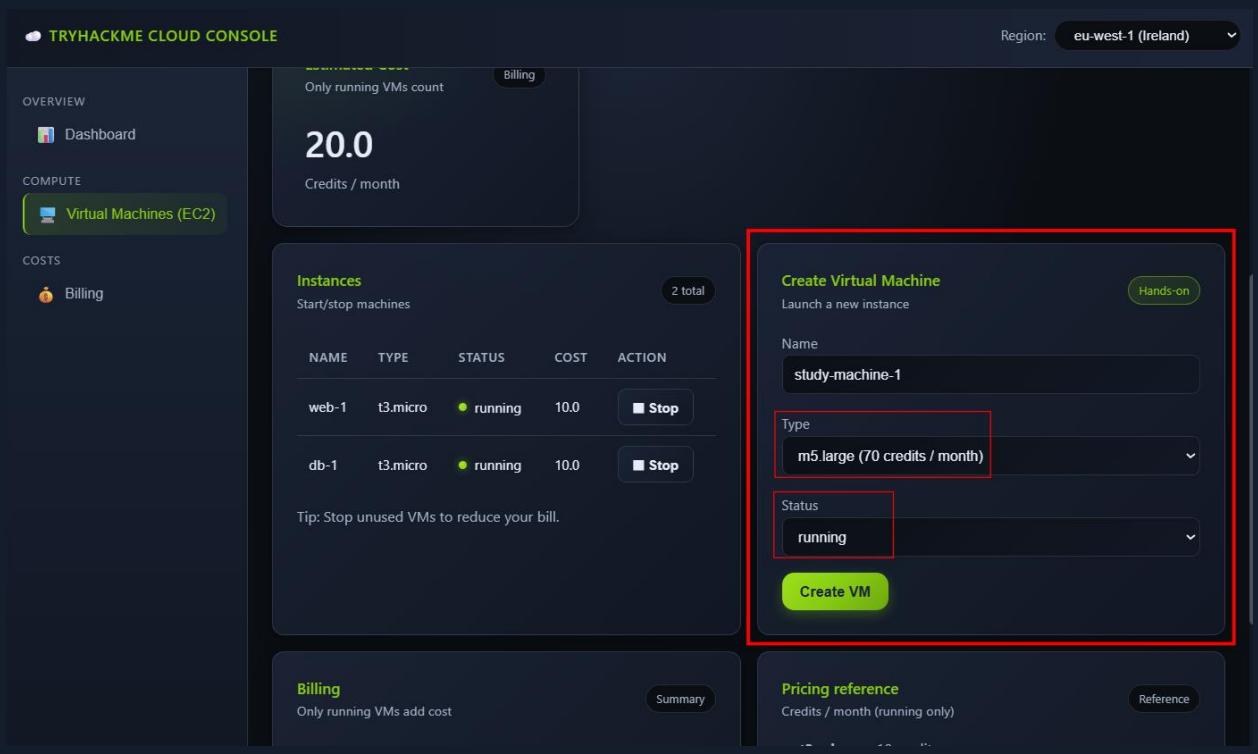
This is the part where you pick a region where your resource will live. The idea is simple. If your users are in Nepal, you want your servers close to Nepal, not on the other side of the world. Closer region means lower latency. For this room Ireland was the closest available option so I chose eu-west-1 Ireland as the region.



[*Fig: Choosing the region for EC2*]

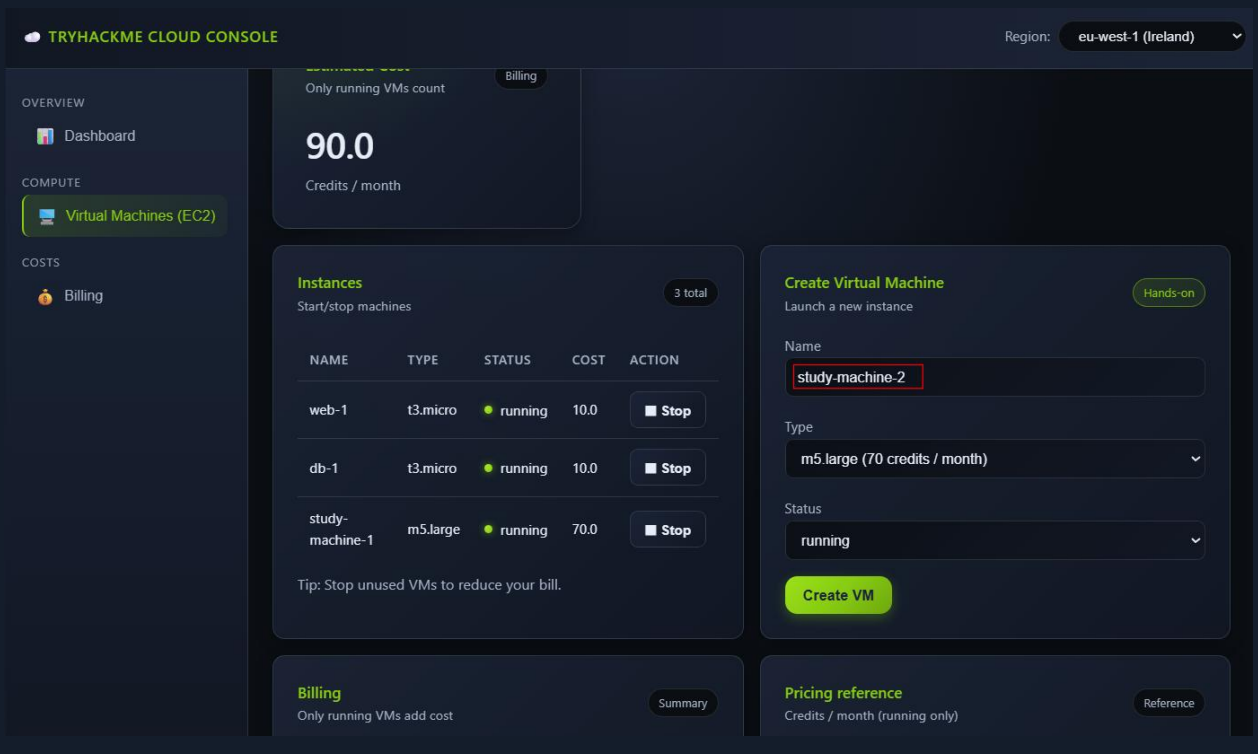
Creating Lab Machin

Now the room takes you to the Create Virtual Machine section where you create two testing computers. I created study-machine-1 first using the m5.large instance type at 70 credits per month.



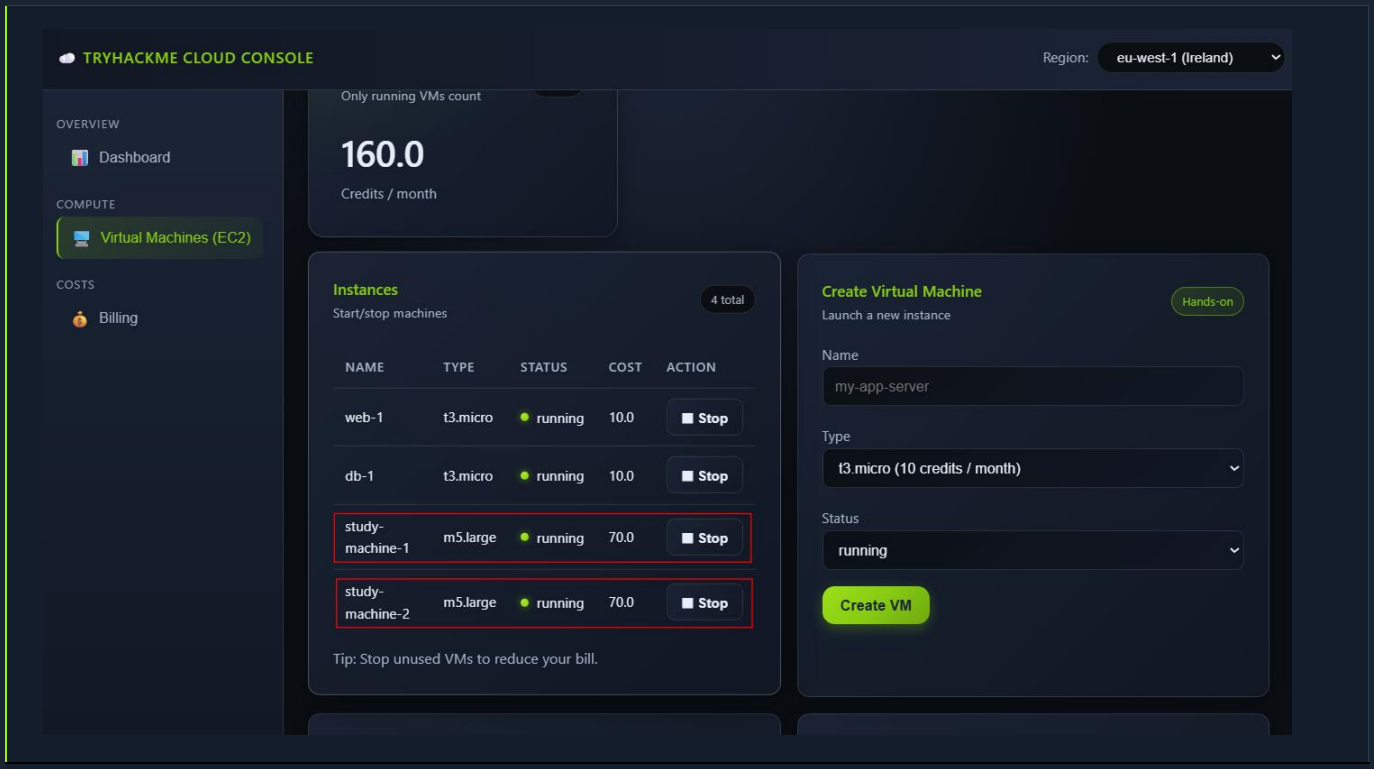
[*Fig: Creating study-machine-1 vm*]

Same process for the second one. I created study-machine-2 with the exact same settings.



[*Fig: Creating study-machine-2 vm*]

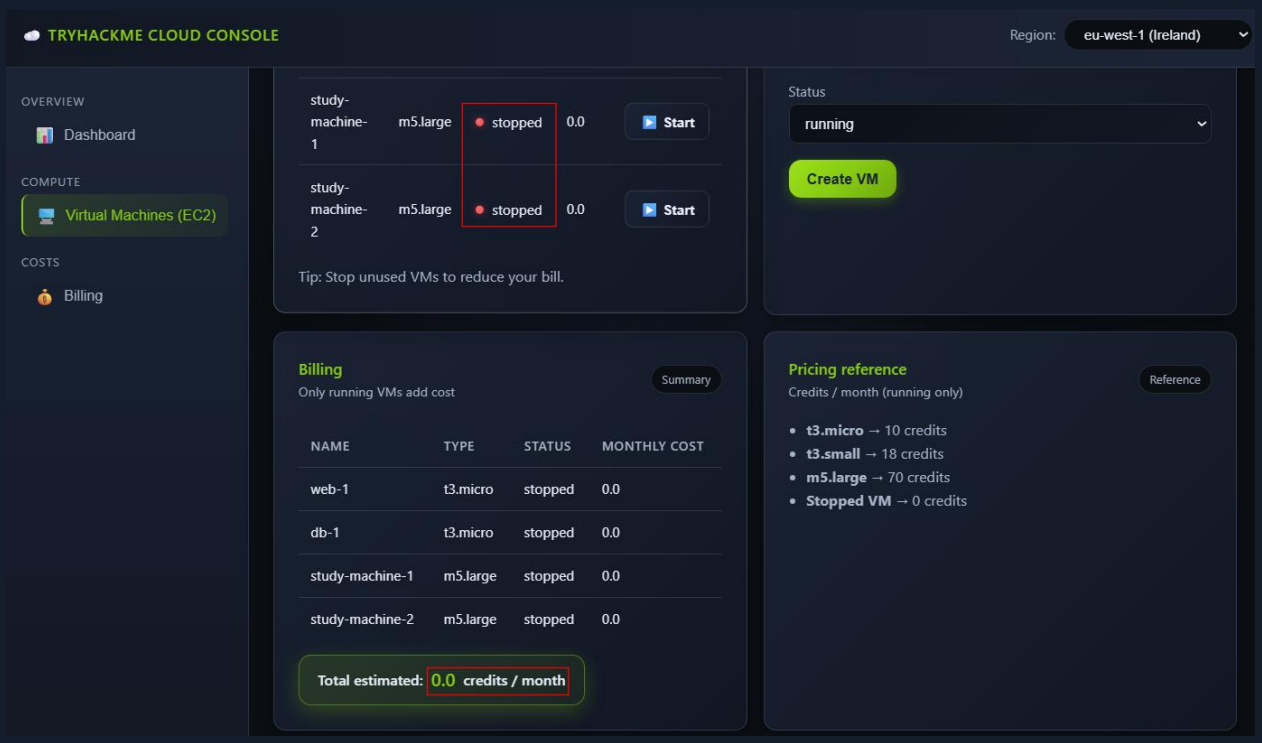
After creating both machines, the instances panel shows all four VMs. web-1 and db-1 were already there from before. Now study-machine-1 and study-machine-2 are added and all four show as running.



[*Fig: study-machine-1 and study-machine-2 both vm running*]

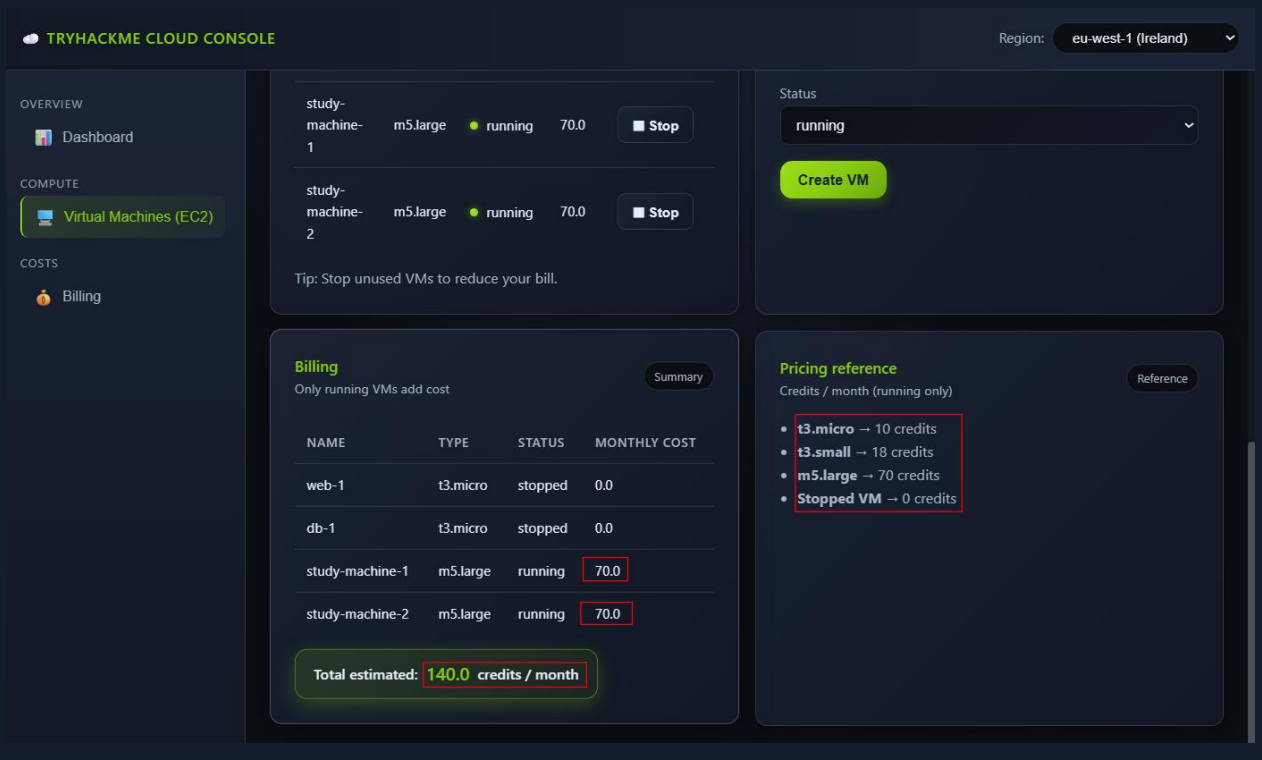
Biling Analysis

Now the room asks you to check the Billing section to understand how credits work. I stopped all the machines and the total estimated cost dropped to zero immediately. Only running VMs cost credits. Stopped VMs cost nothing.



[*Fig: Billing section with total estimated cost by stopping both vm*]

Then I started study-machine-1 and study-machine-2 again to see the running cost.

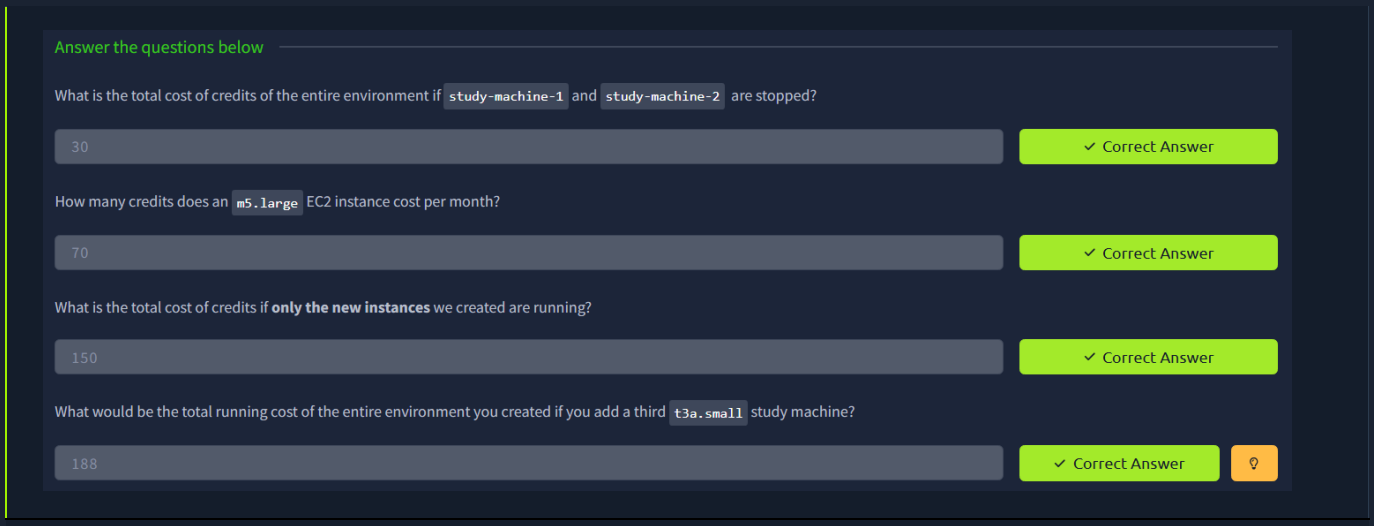


[*Fig: Billing section with total estimated showing 140.0 credits per month*]

> **NOTE**
>
> The billing model here reflects real AWS behaviour. You only pay for what is running. The moment you stop an instance the cost for that instance drops to zero. This is why in production environments people set up auto scaling and scheduled stops. Running an m5.large 24 hours a day when you only need it for 2 hours costs 12 times more than it should.

03 Answering the Questions

The room ends with a knowledge check. Here are the questions and the answers I worked out from the billing data.



[*Fig: Admin Panel showing camaleon CMS version 2.9.0 in footer*]

NOTE

For billing questions like these always work through the pricing reference table first. The room gives you: t3.micro = 10 credits, t3.small = 18 credits, m5.large = 70 credits, stopped VM = 0 credits. Write those down then substitute into each question. It is just arithmetic once you have the numbers.

Summary

#	Phase	What I Learned
1	**Cloud Terminology**	EC2 is a virtual computer in the cloud, instance types define power and cost
2	**Instance Types**	t2, t3, m5 families, size and generation determine the credit cost
3	**Region Selection**	Always pick the region closest to your users for lower latency
4	**Creating VMs**	study-machine-1 and study-machine-2 are both created as m5.large at 70 credits each
5	**Billing Model**	Only running VMs cost credits, stopped VMs cost zero
6	**Knowledge Check**	All four-billing questions answered correctly using the pricing reference table

Main takeaway

The biggest lesson from this room is that cloud billing is not magic. It is simple arithmetic based on what is running and what type it is. The moment I understood that stopped equals zero I understood why every real cloud environment should have auto-stop rules and scheduled shutdowns for anything not needed 24/7.

The second thing is that region choice matters more than people think. Latency affects user experience and sometimes data residency laws also force you into specific regions. It is not just a dropdown you click randomly.

This room is a foundation. Everything in cloud security builds on top of knowing what EC2 is, what instance types exist, and how the billing model works. You cannot attack or defend cloud infrastructure you do not understand.

Hope this helped. Happy learning!